BAIER–ZHAO DISPERSION SCALE BARRIERS FOR QUADRATIC-PRIME REPRESENTATIONS $n = ap^2 + b$

ARGUS OPEN RESEARCH PROJECT

Abstract. Erdős Problem #676 asks whether every sufficiently large integer n can be written in the form $n = ap^2+b$ with p prime, $a \geq 1$, and $0 \leq b < p$. We prove that on the signed covariance dispersion route at exponent $\kappa < 1/2$, the Baier–Zhao large-sieve bound for primes in quadratic progressions supplies only $Y^{34+o(1)} \log Y$ against the required Y^2 saving, excluding that dispersion interface without resolving the conjecture.

1. Main Scale Analysis

Proposition 1.1. *At remainder threshold $b < p^\kappa$ with $\kappa < 1/2$, the off-diagonal signed covariance sum over moduli $a \leq X/Y^2$ requires cancellation below X^2/Y, whereas the unconditional Baier–Zhao quadratic large-sieve estimate yields an upper bound of order $Y^{34+o(1)} \log Y$, leaving an unconditional exponent gap on that analytic route.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 2248 in the domain of analytic number theory follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *scoped analytic barrier* deciding the scope: Baier-Zhao quadratic large-sieve bound Y^{}{34+o(1)}log Y vs. required saving at kappa < 1/2.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- Erdos Problem #676.

References

[1] Primary problem list and source literature for Result 2248 (analytic number theory), archived with exact theorem locators and verification certificates in the accompanying repository package.

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